King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 208 Major Exam II Term 223

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. Let r_1, r_2, r_3 be the roots of the characteristic equation of the differential equation

$$3y''' - 7y'' - 7y' + 3y = 0.$$

If $r_1 \le r_2 \le r_3$, then $r_1^2 + r_2 r_3 =$

(a) $-\frac{5}{3}$

(b) 0

(c) $\frac{10}{9}$

(d) $\frac{7}{3}$

(e) 2

2. Which one of the following statements is True about the following vectors in $\mathbb{R}^4$:

$$v_1 = (3, 0, -4, 1), \quad v_2 = (-3, -6, 4, 2), \quad v_3 = (-3, 2, 4, -2)?$$

(a) v_1, v_2, v_3 are linearly dependent

(b) $v_3 = 5v_1 - v_2$

(c) $v_3 = v_1 + v_2$

(d) v_1, v_2, v_3 are linearly independent

(e) $v_3 = 2v_1 + v_2$

3. Find the rank of the matrix

$$A = \begin{bmatrix} 1 & 1 & 3 & 3 & 1 \\ 2 & 3 & 7 & 8 & 3 \\ 4 & 6 & 14 & 16 & 6 \\ 3 & 1 & 7 & 5 & 4 \end{bmatrix}.$$

(a) 5

(b) 2

(c) 1

(d) 4

(e) 3

4. If $y = Ae^{4x}\cos(3x) + Be^{4x}\sin(3x)$ is the solution of the initial value problem

$$y'' - 8y' + 25y = 0, \quad y(0) = 3, \quad y'(0) = 0,$$

then $AB =$

(a) 10

(b) -12

(c) 0

(d) -9

(e) 6

5. Which pairs of vectors in $\mathbb{R}^3$ are linearly independent?

(a) $v_1 = (2, -1, 3),\ v_2 = (-10, 5, -15)$

(b) $v_1 = (2, -1, 3),\ v_2 = (0, 0, 0)$

(c) $v_1 = (2, -1, 3),\ v_2 = (4, -2, 6)$

(d) $v_1 = (2, -1, 3),\ v_2 = (2, 1, -3)$

(e) $v_1 = (2, -1, 3),\ v_2 = \left(\frac{1}{3}, -\frac{1}{6}, \frac{1}{2}\right)$

6. If y_p is a particular solution of the differential equation

$$y'' + 3y' - 4y = 2x^2,$$

then $y_p(-1) =$

(a) $-\frac{1}{2}$

(b) $\frac{5}{16}$

(c) $\frac{7}{4}$

(d) $-\frac{9}{16}$

(e) $\frac{3}{4}$

7. A particular solution of the differential equation

$$y'' + 16y = 16\sec^2(4x)$$

is

(a) $y_p = -4 + \sin(4x) \cdot \ln|\sec(4x) + \tan(4x)|$

(b) $y_p = -\tan(4x) + \sec(4x)\tan(4x)$

(c) $y_p = -\sec(4x) + \ln|\sec(4x) + \tan(4x)|$

(d) $y_p = -1 + \cos(4x) \cdot \ln|\sec(4x) + \tan(4x)|$

(e) $y_p = -1 + \sin(4x) \cdot \ln|\sec(4x) + \tan(4x)|$

8. Which set of the following functions forms a linearly independent set of functions on $(-\infty, \infty)$?

(a) $\{1,\ 2x,\ x^3\}$

(b) $\{1,\ \cos^2 x,\ \cos(2x)\}$

(c) $\{x,\ 2x^2,\ 3x - x^2\}$

(d) $\{0,\ x,\ e^x\}$

(e) $\{2,\ \sin^2 x,\ \cos^2 x\}$

9. Find a differential equation with constant coefficients whose general solution is

$$y = c_1 e^{3x} + c_2 e^{-x}\sin x + c_3 e^{-x}\cos x.$$

(a) $y''' - 2y'' - y' - 4y = 0$

(b) $y''' - y'' - 4y' - 6y = 0$

(c) $y''' + y'' + 4y' - 6y = 0$

(d) $y''' + 3y'' - 4y' + 6y = 0$

(e) $y''' - y'' - 6y' = 0$

10. Find a basis for the solution space of the system

$$\begin{cases} x_1 - 3x_2 - 9x_3 - 5x_4 = 0 \\ 2x_1 + x_2 - 4x_3 + 11x_4 = 0 \\ x_1 + 3x_2 + 3x_3 + 13x_4 = 0 \end{cases}$$

(a) $\{(4, 3, 0, -1),\ (-20, -15, 0, 5)\}$

(b) $\{(0, 2, -1, 1),\ (3, 1, 0, 4)\}$

(c) $\{(-1, -5, 1, 1)\}$

(d) $\{(3, -2, 1, 0),\ (-4, -3, 0, 1)\}$

(e) $\{(6, -4, 2, 0)\}$

11. Find all values of k for which the following vectors

$$v_1 = (1, 7, -3), \quad v_2 = (2, k, 4), \quad v_3 = (k, 0, k)$$

form a basis for $\mathbb{R}^3$.

(a) $k \neq 0$ and $k \neq -\dfrac{7}{2}$

(b) $k \neq 1$ and $k \neq -4$

(c) $k \neq 0$ and $k \neq -3$

(d) $k \neq 2$

(e) $k \neq -1$

12. Find the Wronskian of the functions $f(x) = e^x$, $g(x) = e^{2x}$, $h(x) = e^{3x}$ on $(-\infty, \infty)$.

(a) $6e^{6x}$

(b) $2e^{6x}$

(c) $18e^{6x}$

(d) $-4e^{6x}$

(e) $-8e^{6x}$

13. Find a general solution of the differential equation

$$y^{(5)} + y^{(4)} - 2y^{(3)} = 0.$$

(a) $y = c_1 + c_2x + c_3e^x + c_4e^{2x}$

(b) $y = c_1 + c_2x + c_3x^2 + c_4e^{-x} + c_5e^{2x}$

(c) $y = c_1 + c_2x + c_3x^2 + c_4e^x + c_5e^{-2x}$

(d) $y = c_1 + c_2x + c_3x^2 + c_4e^x + c_5xe^x$

(e) $y = c_1 + c_2e^x + c_3e^{-2x}$

14. An appropriate form of a particular solution of the differential equation

$$(D + 1)(D - 1)^2(D^2 + 4)y = 1 - 3e^x + 2x\sin(2x)$$

is

(a) $y_p = A + Bx^2e^x + (Cx + D)\sin(2x)$

(b) $y_p = A + Bx^2e^x + (Cx + Dx^2)\cos(2x) + (Ex + Fx^2)\sin(2x)$

(c) $y_p = A + Bxe^x + (Cx + Dx^2)\cos(2x) + (Ex + Fx^2)\sin(2x)$

(d) $y_p = A + Be^x + (Cx + D)\cos(2x) + (Ex + F)\sin(2x)$

(e) $y_p = A + Be^x + (Cx + Dx^2)\sin(2x)$

Answer Key

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Q	Ans	Brief explanation
1	**(e)**	$(r+1)(3r-1)(r-3)=0$: roots $-1, \frac{1}{3}, 3$, so $1+1=2$.
2	**(a)**	$4v_1+v_2+3v_3=\mathbf{0}$, so the vectors are linearly dependent.
3	**(e)**	Row reduction leaves 3 nonzero rows, so the rank is 3.
4	**(b)**	$y(0)=A=3$, $y'(0)=4A+3B=0 \Rightarrow B=-4$, so $AB=-12$.
5	**(d)**	Only $(2,1,-3)$ is not a scalar multiple of $(2,-1,3)$; the other pairs are multiples.
6	**(d)**	$y_p=-\frac{1}{2}x^2-\frac{3}{4}x-\frac{13}{16}$, so $y_p(-1)=-\frac{9}{16}$.
7	**(e)**	$W=4$, $u_1=-\sec 4x$, $u_2=\ln\lvert\sec 4x+\tan 4x\rvert$: $y_p=-1+\sin 4x\ln\lvert\sec 4x+\tan 4x\rvert$.
8	**(a)**	Only $\{1, 2x, x^3\}$ has no function that is a combination of the others.
9	**(b)**	Roots $3, -1\pm i$: $(r-3)(r^2+2r+2)=r^3-r^2-4r-6$.
10	**(d)**	Free $x_3=t$, $x_4=s$: $t(3,-2,1,0)+s(-4,-3,0,1)$.
11	**(a)**	$\det=4k^2+14k=2k(2k+7)\neq 0$, so $k\neq 0,-\frac{7}{2}$.
12	**(b)**	Factor out e^{6x}; the remaining determinant is 2, so $W=2e^{6x}$.
13	**(c)**	$r^3(r+2)(r-1)=0$: $y=c_1+c_2x+c_3x^2+c_4e^x+c_5e^{-2x}$.
14	**(b)**	y_c has $e^x, xe^x, \cos 2x, \sin 2x$, so use x^2e^x and an extra factor x on trig terms.